Contact

www.linkedin.com/in/jdavis-swe (LinkedIn)

Top Skills

Generative AI
Keras
p95

Joel D.

Senior AI Platform Engineer | Applied AI Engineer | Generative AI Engineer | Agentic AI | RAG | LLM | Forward Deployed Engineer
United States

Summary

Senior AI and Software Engineer with 12+ years of experience building production AI systems, scalable backend platforms, and intelligent applications across startups and leading technology companies.Specialized in Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, and AI platform engineering. Experienced in building AI-powered products, enterprise integrations, and cloud-native systems using Python, FastAPI, LangGraph, LlamaIndex, AWS, and Kubernetes.Currently a Founding Software Engineer at CareYaya Health Technologies, leading backend and AI development for healthcare automation and conversational AI products. Previously worked on LLM applications, enterprise AI integrations, intelligent search, recommendation systems, and natural language processing at OpenAI, Airbnb, and Google.Passionate about building practical AI solutions, taking products from MVP to production, and solving complex engineering challenges through scalable architecture, automation, and cross-functional collaboration.

Experience

CareYaya Health Technologies
Founding AI Software Engineer
January 2023 - Present (3 years 10 months)

· Lead backend and AI platform engineering for healthcare applications, building scalable production systems using Python, FastAPI, PostgreSQL, and AWS.

· Developed AI-powered document processing and automation using OpenAI APIs, Amazon Textract, AWS Lambda, and Celery, reducing manual processing effort by up to 62%.

· Designed Retrieval-Augmented Generation (RAG) pipelines using LlamaIndex and pgvector, and agentic workflows with LangGraph and human-in-the-loop approvals.

· Built and optimized cloud-native services using Amazon ECS Fargate, Redis, and Amazon SQS, improving p95 latency by up to 28%.
· Collaborated on MLOps and data infrastructure using Databricks, PySpark, MLflow, and Ragas to support reliable AI evaluation and deployment.

OpenAI
Senior AI Software Engineer
March 2020 - December 2022 (2 years 10 months)

· Developed GPT-3-powered enterprise AI applications for document processing, knowledge retrieval, and workflow automation using Python, Django REST Framework, and OpenAI APIs.
· Built semantic search and retrieval solutions using OpenAI embeddings, Elasticsearch, FAISS, and GPT-3 to improve access to enterprise knowledge.
· Designed reusable backend services and API integrations using PostgreSQL, Redis, Celery, Docker, and Kubernetes, supporting scalable LLM-powered applications.
· Implemented AI-powered request classification and automation workflows, reducing manual triage work by up to 36% in partner pilots.
· Collaborated with research, product, and engineering teams on model evaluation, API reliability, and production-ready AI integrations.

Airbnb
Software Engineer
May 2017 - February 2020 (2 years 10 months)

Airbnb was using AI to better understand property listings and improve how homes were presented to guests. I worked on computer vision systems for room classification, amenity recognition, and large-scale listing photo analysis. Helped turn image intelligence into useful metadata for listing galleries and host workflows.

Google
Software Engineer
June 2014 - April 2017 (2 years 11 months)

Google was advancing conversational AI toward more natural and context-aware interactions. I contributed to language-understanding systems for intent, entities, context, and follow-up questions, while connecting user requests with services like Search, Maps, and Calendar. Focused on making conversational experiences more useful and action-oriented.

Education

University of Miami

Bachelor's Degree, Computer Science · (September 2010 - May 2014)